



GHOST FONT DECODER

MOTION-AWARE TEXT RECOVERY

GHOST FONT DECODER

THE MESSAGE IS THE MOTION.



COMPLETE PROJECT HANDBOOK

Technical Reference and Operating Guide

Pipeline, evidence, deployment, risks, maintenance, and verified execution records.

CORE PRINCIPLE The message is encoded in motion, not in any single frame.

Preface

A permanent record of the project's purpose, design, evidence, and operating decisions.

Why this reference exists

Ghost Font Decoder combines computer vision, agent integrations, and strict review rules. This document preserves the complete proposal and technical knowledge so future owners do not have to rely on memory.

Contents

Preface	2
1. Executive proposal	4
Recommendation	4
Expected business value	4
2. The problem in simple terms	5
Why still-image analysis fails	5
3. Pipeline at a glance	6
4. Detailed pipeline - signal extraction	7
5. Detailed pipeline - reconstruction	8
6. Outputs and how to interpret them	9
Review checklist	9
7. Architecture and deployment choices	10
Repository packaging	10
8. Reference evidence - real agent executions	11
9. Reference evidence - prompt and plugin paths	12
10. Operating guide	13
Install and verify	13
Run the included sample	13
Run a new authorized video	13
11. CLI parameters and decision rules	14
Decision tree	14
12. Failure modes and troubleshooting	15
13. Security, privacy, and limitations	16

Security meaning 16

Known limitations 16

14. Proposed adoption plan **17**

Pilot acceptance criteria 17

Ownership 17

15. One-year return checklist **18**

16. Appendix - glossary, files, and references **19**

Plain-language glossary 19

Authoritative project references 19

External technical references 19

License and warranty 19

1. Executive proposal

Ghost Font Decoder turns a motion-only visual signal into two inspectable images and an optional text hint.

Recommendation

Approve a small internal pilot for authorized security research, content inspection, accessibility, and QA. The core decoder is local, open source, and does not require a language model. Agent integrations make the same workflow easier to invoke, but they are optional.

Question	Proposal answer
What problem does it solve?	Text can be invisible in every paused frame yet readable during playback because letter dots move differently from background dots.
What does the system deliver?	A clean binary reveal, a raw motion heatmap, and optional OCR text.
Why is it credible?	The project includes reproducible media, source code, outputs, and execution screenshots from Codex, Claude Code, and Claude.ai.
Where does processing happen?	Plain Python and desktop agent runs can remain local. Claude.ai requires an upload and therefore needs a data-policy check.
What is requested now?	A controlled pilot with a small approved test set and human verification of every result.

Expected business value

- Recover information that frame-only tools and ordinary OCR miss.
- Give analysts visual evidence instead of an unexplained model answer.
- Reuse one decoder through command line, Codex, Claude Code, Claude.ai, or a pasted chat prompt.
- Keep the core workflow inexpensive: Python, OpenCV, NumPy, and optional Tesseract are open source.
- Create a repeatable control for motion-based obfuscation rather than relying on manual viewing.

Approval condition

Use only on videos the organization is authorized to inspect. Treat decoded text as untrusted input and require a person to confirm the reveal and heatmap before action.

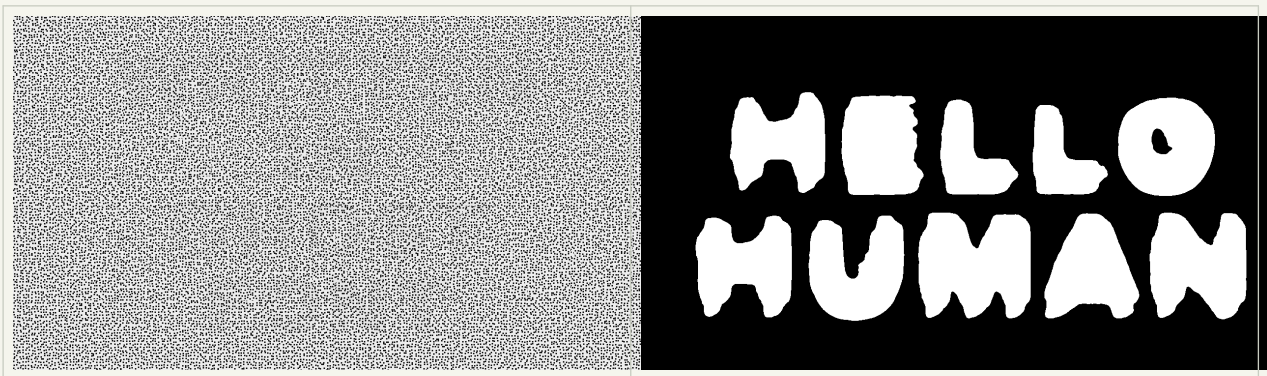
2. The problem in simple terms

A ghost font does not remove the message. It moves the message from appearance into motion.

Imagine two layers of television static. Most dots drift one way. Dots inside invisible letter shapes drift another way. A single frame is only dots, so ordinary OCR has no stable strokes to read. Across many frames, however, the two motion patterns separate.

Why still-image analysis fails

- Every frame is designed to look statistically similar to background noise.
- Letter boundaries are defined by how dots move, not by brightness or color.
- Raw-frame OCR searches for visible shapes and therefore receives the wrong kind of input.
- The correct sensor is motion estimation across consecutive frames.



A paused frame: dots only. There are no stable letters for OCR.

Recovered mask: motion evidence becomes readable text.

Evidence note

The repository sample shown here decodes to HELLO HUMAN. The separate real-agent screenshots later in this proposal show a different test clip decoding to I LOVE YOU.

3. Pipeline at a glance

The decoder converts video into a stable score map, cleans that map, and saves two human-checkable outputs.

Stage	Plain-language purpose	Main implementation
1. Read frames	Turn the video into a sequence of grayscale pictures.	OpenCV VideoCapture and BGR-to-gray conversion
2. Measure motion	Estimate where every pixel moved between two pictures.	DIS optical flow by default; Farneback is optional
3. Find background drift	Use the movement shared by most pixels as the background motion.	Median horizontal and vertical flow
4. Score opposite motion	Make letter pixels bright when they move against the background.	Residual flow projected onto the anti-background direction
5. Register text drift	Keep the letter region aligned while it slowly moves across the clip.	Gaussian smoothing, phase correlation, affine translation
6. Accumulate evidence	Add weak clues from many frame pairs until letters become strong.	Registered score-map summation
7. Build clean mask	Separate likely letters from the remaining noise.	99.5th percentile normalization, Otsu threshold, morphology
8. Recover faint glyphs	Bring back small real letters that fell below the main threshold.	Compact connected-component checks
9. Remove artifacts	Discard tiny specks and wide edge streaks.	Area, aspect-ratio, and edge-position filters
10. Save and read	Crop, enlarge, save both views, and optionally ask OCR for a hint.	revealed.png, revealed_heatmap.png, optional Tesseract

The governing rule

The visual outputs are the evidence. OCR is optional and never authoritative. Read revealed.png directly and use revealed_heatmap.png to confirm weak or merged shapes.

4. Detailed pipeline - signal extraction

Steps 1 through 5 turn a video into aligned per-pixel motion evidence.

01

Open the video and select frames

In simple terms: Read the clip from beginning to end. Use every frame unless stride asks for every second, third, or later frame.

What the code does: `iter_gray_frames` opens the video, converts each selected frame to grayscale, honors `--stride` and `--max-frames`, and stops with a clear error if the file cannot be opened.

Result: A stream of equally sized grayscale frames.

02

Compute dense optical flow

In simple terms: Draw a tiny motion arrow for every pixel between the previous frame and the current frame.

What the code does: `DISOpticalFlow` with the MEDIUM preset is the default. Farneback uses a multi-level pyramid and can be selected with `--method farneback`.

Result: A two-channel motion field containing horizontal and vertical displacement.

03

Estimate the background motion

In simple terms: Assume most pixels belong to the background, then use their shared movement as the camera or background drift.

What the code does: The median of all flow vectors is calculated for each frame pair. A median is resistant to the smaller text region.

Result: One background-motion vector per frame pair.

04

Score motion that opposes the background

In simple terms: Pixels moving against the dominant direction are likely to belong to a letter.

What the code does: The background vector is subtracted from every flow vector. Positive projection onto the opposite direction becomes the pair score. If background motion is nearly zero, residual magnitude is used instead.

Result: A bright score map for that frame pair.

05

Track the drifting letter region

In simple terms: Hold the letters still on an imaginary sheet before adding evidence from the next moment.

What the code does: The score map is blurred with a 31 by 31 Gaussian kernel. Phase correlation estimates translation. Updates are accepted only when response exceeds 0.05 and motion is under 30 pixels.

Result: A cumulative text-drift estimate and a registered score map.

5. Detailed pipeline - reconstruction

Steps 6 through 10 turn accumulated evidence into readable and reviewable output.

06

Accumulate evidence across time

In simple terms: Add many weak motion clues together. Real letter motion repeats; unrelated noise tends to cancel or remain weak.

What the code does: Each pair score is translated back by the cumulative drift and added into one floating-point score image.

Result: A high-energy motion image in which glyph regions are stronger than background noise.

07

Normalize and choose a threshold

In simple terms: Convert large numeric scores into visible brightness, then choose a data-driven boundary between foreground and background.

What the code does: Scores are clipped at zero, scaled using the 99.5th percentile, blurred with a 5 by 5 Gaussian kernel, and thresholded with Otsu's method.

Result: A first binary mask and a normalized heatmap.

08

Close gaps and recover faint glyphs

In simple terms: Join broken strokes, remove isolated grains, and rescue small letters that are consistently brighter than their surroundings.

What the code does: A 7 by 7 elliptical close/open cleans the strong mask. A half-Otsu faint threshold and connected-component tests recover compact, coherent components with sufficient peak energy.

Result: A more complete mask that can preserve a small isolated letter such as I.

09

Reject implausible artifacts

In simple terms: Remove shapes that look like processing streaks rather than letters.

What the code does: Components below a size floor are deleted. Very wide, short bands are rejected when they touch an edge or span at least one third of the frame.

Result: A cleaner letter mask with fewer camera-pan and registration bands.

10

Crop, enlarge, save, and optionally OCR

In simple terms: Make the message easy to inspect and save both the clean answer view and the raw evidence view.

What the code does: The mask defines a padded crop shared by both outputs. Small crops are enlarged up to a 1000-pixel long side, with a maximum 4x scale. Optional Tesseract OCR runs on the full mask and includes a check for an isolated uppercase I.

Result: revealed.png, revealed_heatmap.png, and an optional console text hint.

6. Outputs and how to interpret them

The two files answer different questions and should always be reviewed together.

	
revealed.png	revealed_heatmap.png
Black background with solid white letter regions. This is the easiest view for a person and the intended OCR input.	Grayscale strength map before final thresholding. It preserves faint structure and helps confirm a weak or merged glyph.

Review checklist

- Confirm that the clean mask contains connected shapes arranged like text.
- Read the mask directly. Do not accept an OCR string that disagrees with visible shapes.
- Use the heatmap to confirm uncertain strokes, especially small top-line characters and rounded letters.
- If the mask is only specks and the heatmap is uniformly dark, report that no text was recovered.
- Store the input filename, command, decoder version, outputs, and reviewer decision together for auditability.

Important

A soft, rounded, or blobby rendering is normal. Do not keep reprocessing a readable result until it looks like a conventional font; that increases the risk of inventing structure from noise.

7. Architecture and deployment choices

The computer-vision core is the same; only the invocation surface and data boundary change.

Option	Best use	Data boundary	Operational note
Plain Python	Production jobs, repeatable testing, controlled environments	Can remain fully local	Most predictable and easiest to pin, log, and automate
Codex skill	Developers and analysts working in Codex	Local task environment, subject to organization setup	Runs decoder, displays both images, and reports text
Claude Code plugin	Repository-centered developer workflow	Local command environment, subject to organization setup	Uses /ghost-decode or a natural-language request
Claude.ai skill	Convenient web or desktop analysis	Video is uploaded to the Claude.ai environment	Requires a policy review and code execution capability
Portable chat prompt	One-off supported chat environments	Depends on the chosen chat service	No plugin install, but runtime behavior is less controlled

Repository packaging

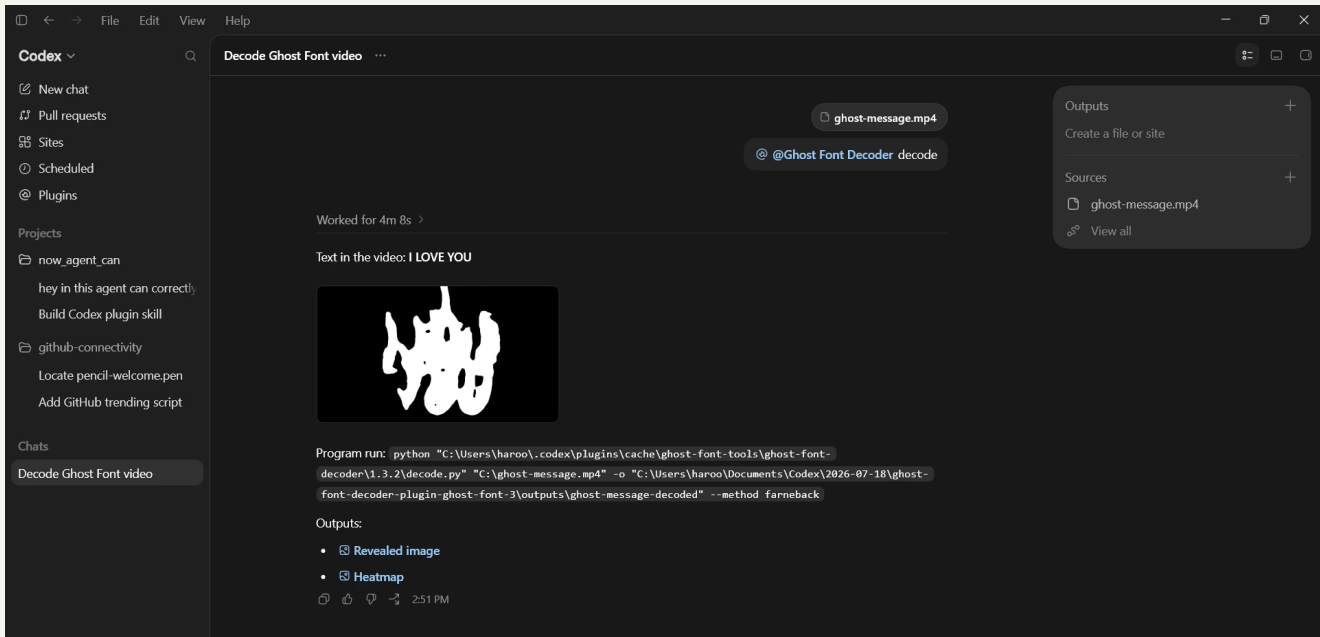
Location	Purpose
decode.py	Authoritative standalone decoder and CLI.
codex-skills/ghost-decode/	Codex skill instructions and interface metadata.
claude-skills/ghost-decode/	Claude Code skill instructions.
commands/ghost-decode.md	Claude Code slash-command wrapper.
claude-ai-skill/	Self-contained uploadable Claude.ai skill source and zip builder.
prompts/decode-in-chat.md	Portable in-chat workflow for supported runtimes.
examples/	Sample video, paused frame, clean reveal, and heatmap.
docs/	Project website, technical documentation, and proof assets.

Production recommendation

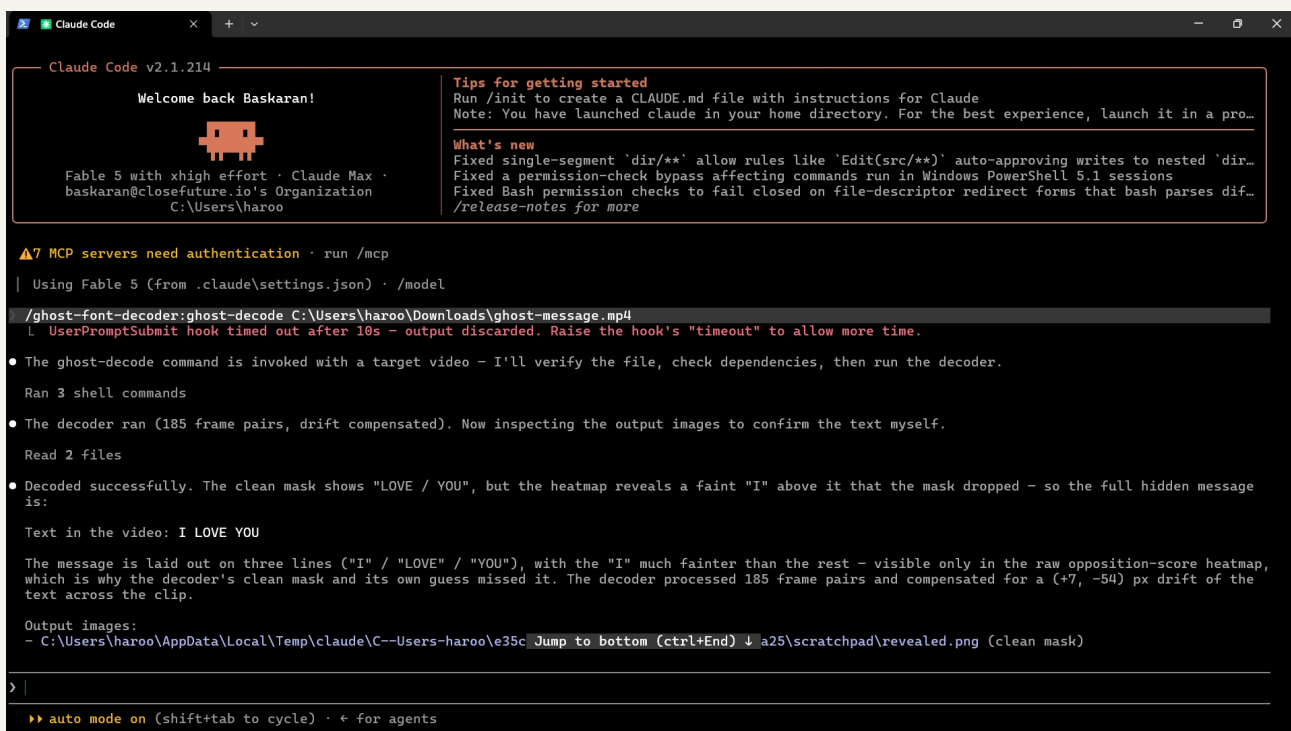
Use the plain Python path as the controlled system of record. Treat agent skills as convenient front ends that call the same decoder and present its evidence.

8. Reference evidence - real agent executions

These images are execution records, not mockups. They show the invocation, reported message, and generated evidence.



Reference 1. Codex invoked the installed decoder skill, displayed outputs, and reported I LOVE YOU.



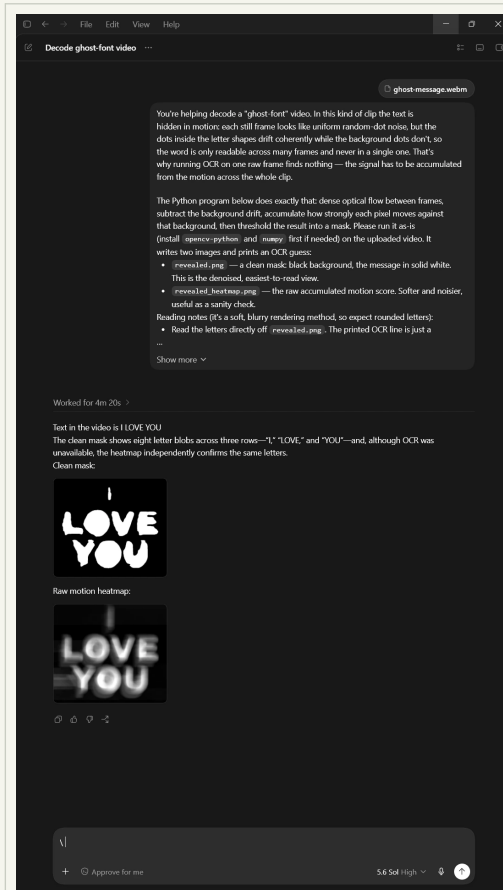
Reference 2. Claude Code processed 185 frame pairs and reported the same recovered message.

What this proves

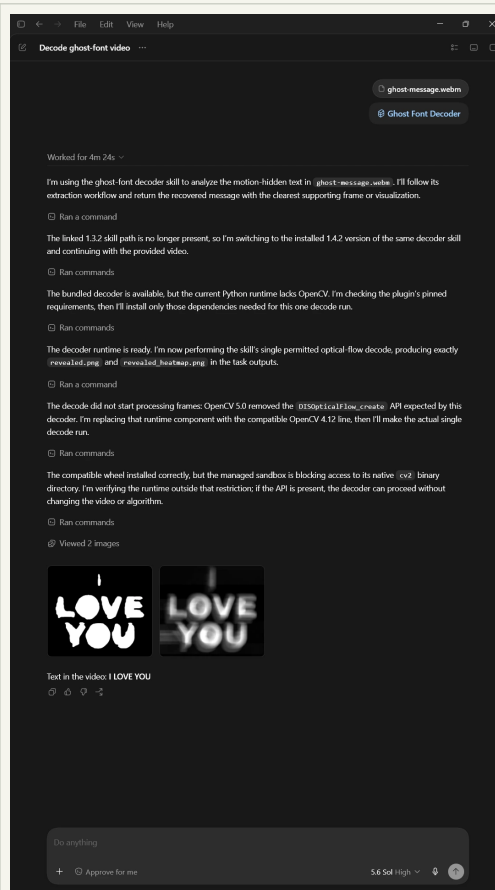
Independent agent surfaces can run the computer-vision workflow and return the same human-readable result. The proof is strongest when the saved reveal and heatmap are retained with the chat record.

9. Reference evidence - prompt and plugin paths

The project supports both a reusable prompt and installed skill or plugin flows.



Codex with the reusable chat prompt.



Codex with the installed Ghost Font Decoder skill.

The prompt path is useful when installation is impossible. The installed skill is better for repeatability because it standardizes the command, output names, and required visual verification. Both paths must preserve the rule: run the decoder, inspect both images, then report the literal text.

10. Operating guide

This section is the shortest reliable path from a new machine to a verified result.

Install and verify

Run these commands from the repository root:

```
python -m venv .venv
.venv\Scripts\activate
pip install -r requirements.txt
python -c "import cv2, numpy; print(cv2.__version__, hasattr(cv2, 'DISOpticalFlow_create'))"
```

Windows compatibility check

The decoder requires `cv2.DISOpticalFlow_create` for the default method. A prior successful Windows run used OpenCV 4.12.0.88. An incompatible OpenCV 5/base installation lacked this API. Verify the attribute before a production run and pin a tested build in the deployment environment.

Run the included sample

```
python decode.py examples/ghost-message.mp4 -o outputs
```

Expected deliverables are `outputs/revealed.png` and `outputs/revealed_heatmap.png`. The repository sample should visually read HELLO HUMAN.

Run a new authorized video

```
python decode.py path\to\video.webm -o outputs\case-001
```

- Open `revealed.png` and read the letters directly.
- Open `revealed_heatmap.png` and confirm uncertain shapes.
- Record the command and preserve both outputs.
- Use OCR only as a hint. If OCR disagrees with the images, trust the images and mark ambiguity.

11. CLI parameters and decision rules

Most clips should use defaults. Change one option only when a specific symptom justifies it.

Argument	Default	Meaning	When to change it
video	required	Path to MP4, MOV, AVI, or WebM input.	Choose one authorized source clip.
-o / --out-dir	.	Directory for the two output images.	Always use a case-specific folder in operational work.
--method	dis	Dense optical-flow engine: dis or farneback.	Use farneback only for a weak or empty result or when DIS is unavailable.
--stride	1	Process every Nth frame.	Try 2 for high-frame-rate material or when adjacent frames move too little.
--max-frames	none	Stop after a fixed number of selected frames.	Use about 200 for a long clip when a few seconds contain enough evidence.
--no-ocr	off	Skip Tesseract completely.	Use when OCR is unavailable, not approved, or unnecessary.

Decision tree

Observed result	Next action
Readable clean mask	Stop. Confirm with heatmap and report the literal text.
Soft or rounded but readable	Stop. This appearance is expected.
Weak or nearly empty	Retry the same decoder once with --method farneback. Add --stride 2 only for high-frame-rate input.
Only specks in mask and dark heatmap	Report that no text was recovered. Do not invent a message.
OCR is empty or wrong	Ignore it and read the mask directly. OCR is optional.
More than one plausible input file	Ask the operator to identify the intended video before running.

12. Failure modes and troubleshooting

Symptoms should lead to bounded, explainable actions rather than endless reprocessing.

Symptom	Likely cause	Safe response
Cannot open video	Bad path, unsupported codec, permission issue, or damaged file.	Confirm path and playback, then convert to a common MP4/WebM codec if authorized.
Fewer than two usable frames	Clip is empty, truncated, or max-frames/stride is invalid for its length.	Use a valid multi-frame video and sensible sampling.
DISOpticalFlow_create missing	OpenCV build does not expose the required DIS API.	Verify cv2 capability. Pin a tested OpenCV 4.12 build or use <code>--method farneback</code> as the explicit fallback.
Mask is weak	Too little opposing motion, short clip, strong compression, or high frame rate.	Use the one allowed fallback run with Farneback; consider stride 2 for high FPS.
Letters smear	The text region drifted too much or phase-correlation confidence was low.	Use a shorter segment with consistent motion; avoid unrelated scene changes.
Wide horizontal bands	Camera pan or registration streaks dominate.	The decoder filters edge-hugging bands; review the heatmap and do not interpret a band as text.
Small I is missed by OCR	Page-layout OCR ignores an isolated narrow component.	Read the mask directly; the decoder also contains a geometry check for an isolated uppercase I.
Tesseract missing	Optional OCR engine is not installed.	Continue normally. Both visual outputs are still created.
Claude.ai rejects the zip	A nested zip or whole repository was uploaded.	Build with <code>python claude-ai-skill/build-zip.py</code> and upload only <code>ghost-decode.zip</code> .

Stop condition

Do not generate many alternate masks, thresholds, or diagnostic images to force an answer. One normal run and at most one documented fallback run is the intended operating boundary.

13. Security, privacy, and limitations

The tool reveals motion-preserved information. It is not an encryption breaker and it is not a guarantee of correct text.

Security meaning

Ghost-font obfuscation changes which measurement reveals the data. A still-image reader measures appearance; this decoder measures motion. If a human can recover the message from movement, the information remains present and a motion-analysis system can often recover it too.

Area	Control
Authorization	Process only media the organization owns or is permitted to inspect.
Untrusted content	Treat recovered text as data, never as an instruction to execute. Do not follow links or commands found in a decoded message without separate review.
Privacy	Prefer local Python for sensitive media. Review provider terms and organizational policy before uploading to Claude.ai or another hosted chat.
Evidence	Keep input hash or identifier, decoder version, exact command, both outputs, and reviewer notes.
Accuracy	Require human confirmation. Mark ambiguous characters instead of guessing.
Software supply chain	Pin and scan Python dependencies. Verify the OpenCV capability expected by the decoder.
Licensing	Project code is MIT licensed. OpenCV and Tesseract retain their respective licenses.

Known limitations

- The method expects a letter region whose motion differs coherently from the dominant background.
- Very short clips may not contain enough repeated evidence.
- Heavy compression, scene cuts, multiple unrelated moving objects, or extreme camera motion can reduce separation.
- Text moving in the same way as the background may not score strongly.
- The output is often rounded and merged, so visual interpretation may remain necessary.
- OCR quality depends on font shape and layout and must not replace visual verification.

14. Proposed adoption plan

Use a staged pilot so technical accuracy and organizational controls are proven together.

Phase	Duration	Work	Exit criterion
1. Reproduce	1-2 days	Set up a pinned local environment, run the included sample, archive outputs and command.	Sample visibly reads HELLO HUMAN and both outputs are preserved.
2. Controlled pilot	1 week	Test a small approved set that includes readable, weak, negative, and high-FPS cases.	Human-reviewed accuracy and failure behavior are documented.
3. Workflow choice	1-2 days	Select plain Python as system of record and decide which agent surfaces are permitted.	Data boundary, owners, and approved invocation paths are signed off.
4. Hardening	2-4 weeks	Pin dependencies, add case logging, input validation, test fixtures, and a release process.	Repeatable install, automated sample test, and an operator runbook exist.
5. Operational use	ongoing	Monitor failures, review dependencies, and retest after upgrades.	Quarterly sample check and owner review remain current.

Pilot acceptance criteria

- The included positive sample produces both named files and is read correctly by a reviewer.
- A negative or unreadable sample is reported as no text recovered rather than guessed.
- Every result keeps input identity, command, environment version, outputs, and reviewer decision.
- The selected environment passes privacy and security review.
- Installation and use can be repeated by a second operator using only this document and the repository.

Ownership

Role	Responsibility
Business sponsor	Approves use cases, data scope, and success criteria.
Technical owner	Maintains decoder, dependency pins, tests, and releases.
Security or privacy reviewer	Approves processing location and evidence retention.
Operator	Runs the approved command and reviews both images.
Second reviewer	Confirms ambiguous or high-impact interpretations.

15. One-year return checklist

Follow this sequence before touching a new case. It is designed for someone who remembers almost nothing about the project.

Step	Action	Why
1	Open the repository	Read README.md and this PDF. Confirm that decode.py and examples/ still exist.
2	Check source control	Review recent changes, release notes, and the last known working dependency versions.
3	Create a clean environment	Use a fresh virtual environment. Do not reuse an unknown global Python installation.
4	Install and inspect versions	Install requirements and record Python, NumPy, OpenCV, and Tesseract versions.
5	Verify DIS support	Run <code>hasattr(cv2, 'DISOpticalFlow_create')</code> . If false, use a tested 4.12 build or the explicit Farneback fallback.
6	Run the sample	Decode <code>examples/ghost-message.mp4</code> . Expect exactly <code>revealed.png</code> and <code>revealed_heatmap.png</code> .
7	Read the sample visually	The sample should read HELLO HUMAN. If it does not, stop and repair the environment.
8	Confirm authorization	Make sure the new video is approved for inspection and its processing location is allowed.
9	Run one normal decode	Use defaults and a new case-specific output directory.
10	Inspect both outputs	Read the clean mask and use the heatmap to confirm uncertain glyphs.
11	Use at most one fallback	Only if weak or empty: retry with Farneback and optionally stride 2 for high FPS.
12	Record and close	Save command, versions, two outputs, literal result, ambiguity notes, and reviewer name.

Memory anchor

The message is not in a frame. The message is in the difference between movements across frames. Motion first, evidence second, OCR last.

16. Appendix - glossary, files, and references

Use these definitions and source pointers when maintaining or explaining the system.

Plain-language glossary

Term	Meaning
Dense optical flow	A motion arrow for every pixel between two frames.
Background flow	The movement shared by most pixels, estimated with the median.
Residual flow	Motion left after subtracting background movement.
Phase correlation	A frequency-based way to estimate how far one image shifted relative to another.
Registration	Moving evidence maps back into the same coordinate system before adding them.
Accumulation	Summing repeated weak evidence over time.
Otsu threshold	An automatic split between darker background scores and brighter foreground scores.
Morphology	Shape cleanup that joins small gaps and removes isolated grains.
Connected component	A group of touching foreground pixels treated as one candidate shape.
OCR	Software that turns visible text shapes into character strings. Here it is optional.

Authoritative project references

- Repository source and README: <https://github.com/haroontrailblazer/ghost-font-decoder>
- Decoder implementation: `decode.py`
- Codex operating rules: `codex-skills/ghost-decode/SKILL.md`
- Claude Code operating rules: `claude-skills/ghost-decode/SKILL.md`
- Claude.ai build and install guide: `claude-ai-skill/README.md`
- Portable workflow: `prompts/decode-in-chat.md`
- Technical website: `docs/docs.html`
- Reproducible assets: `examples/` and `docs/assets/proof/`

External technical references

- OpenCV DIS optical flow class reference: https://docs.opencv.org/master/de/d4f/classcv_1_1DISOpticalFlow.html
- OpenCV optical-flow tracking reference, including Farneback: https://docs.opencv.org/master/dc/d6b/group__video__track.html
- OpenCV phaseCorrelate namespace reference: <https://docs.opencv.org/master/d2/d75/namespacecv.html>
- Tesseract user manual: <https://tesseract-ocr.github.io/tessdoc/>

License and warranty

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Final takeaway

Ghost fonts are an obfuscation technique, not encryption. This project provides a transparent, evidence-producing way to inspect the motion signal while keeping a human reviewer in control.